Chapter 17

When and how to replace missing teeth

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DO WE NEED TO REPLACE MISSING TEETH?

The loss of any tooth does not automatically mean the need for a prosthetic replacement. Patients should be aware that tooth replacement has a biological and financial cost with a long-term commitment to maintenance. Complications can arise with any restorative intervention and a sound justification for tooth replacement is required. Patients should always be informed of the alternatives, irrespective of cost, for tooth replacement and they should be informed of the potential advantages and disadvantages to each. It is only then that informed consent can be given by the patient.

WHEN TO REPLACE MISSING TEETH?

Appearance

Tooth loss can have a devastating effect on a patient's appearance, self-confidence and motivation. The loss of an upper central incisor is an obvious indication for prosthodontic replacement (Figure 17.1), but the technical challenges of designing the restoration must be taken together with the impact of the 'smile line'. This assessment should be undertaken when the patient is talking and smiling in a relaxed environment and not from under forced conditions! For some patients the replacement of an upper premolar tooth may be unnecessary, but for those with a wide and high smile line tooth replacement may be justified. Whatever the clinical situation, it is advisable that tooth replacement based on appearance should be led by the patient and not the dentist.

Occlusal stability

When a tooth is missing the occlusion can change, with overeruption, tilting or drifting of opposing or adjacent

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Figure 17.1 A patient with a missing upper left central incisor tooth which poses a major aesthetic problem. The patient also has a high and wide smile line involving both first and second premolar teeth.

teeth, causing a condition commonly referred to as an unstable occlusion. Consider the 25-year-old patient in Figure 17.2. The loss of the upper first and second premolar and lower first and second molar teeth has led to an unstable occlusion. The failure to replace these missing teeth led to overeruption of the lower second premolar and upper first and second molar teeth into the opposing edentulous spaces. This now complicates prosthodontic replacement as there is inadequate space between the teeth and opposing edentulous ridge.

The overeruption of teeth into edentulous spaces does not always occur and is impossible to predict. Therefore, if restoration of occlusal stability is the only clinical indication for tooth replacement following an extraction, it is advisable to monitor tooth movement over time. This tooth movement or dento-alveolar compensation can occur quickly in some patients and the tooth movement seen in the patient in Figure 17.2 occurred after 6 months. To avoid such a clinical situation, review appointments should be advised and a suggested timing would be after 3 months in the first instance. Baseline study models taken soon after the extraction may also be useful in confirming any tooth movement.



Figure 17.2 A patient with an unstable occlusion and overeruption of the lower premolar and upper molar teeth. Note the alveolar tissues have accompanied the tooth movement, hence the term dento-alveolar compensation.

Function

Mastication

Patients frequently complain of difficulty in eating and trauma to the edentulous ridges following tooth loss. The latter problem can only be addressed with a prosthodontic replacement but the former is subjective and individual to each patient. There is little evidence to confirm that loss of molar teeth is accompanied by a significant loss of masticatory function. To the contrary, there is a large body of evidence to support the concept of the shortened dental arch (SDA) (Kanno & Carlsson, 2006). The posterior teeth are mainly involved in mastication and the SDA was proposed as a treatment option if molar teeth had been extracted. The minimum number of occluding pairs of teeth was the four premolar teeth or two pairs of occluding molar teeth. Today, the SDA is generally accepted as a premolar to premolar occlusion, following the loss of the molar teeth.

Opponents to the concept of the SDA have cited loss of masticatory function, mandibular displacement and problems such as temporomandibular dysfunction (TMD). There is some evidence to suggest that these complications are more likely in younger patients and those with unilateral tooth loss. In addition, some patients have reported difficulty in chewing harder foods though they tend to modify their diet accordingly. The other important factor in assessing the role of the SDA is the challenges associated with the prosthodontic replacement of molar teeth. For many patients replacing molar teeth with implants is prohibitively costly and the alternative of removable prostheses, particularly in the mandible with free-end saddles, means denture-based solutions are often not successful. The strength of evidence strongly supports the concept of adopting the SDA when the molar teeth have been extracted.

Does the SDA lead to increased wear of remaining teeth?

It would seem logical that if patients have missing molar teeth and a lack of 'posterior support', there would be increased tooth wear of the remaining teeth as these take the entire functional load. This is a frequent comment and perhaps misconception of many dentists. In those patients with tooth wear the rate generally increases over a lifetime but those without evidence of wear are unlikely to develop wear following the loss of the teeth.



Figure 17.3 A patient with missing posterior teeth. The remaining anterior teeth are worn, mainly as a result of erosion (cupping out of the incisal edges is seen) and not lack of 'posterior support'.

The patient in Figure 17.3 has lost all their posterior teeth and due to the severity of posterior alveolar ridge resorption is unable to wear a successful removable prosthesis. The lower anterior teeth are worn, with cupping of the incisal edges resulting from erosion, though attrition might be a contributing factor to the tooth wear. Whether the tooth wear would have reached the same severity if the posterior teeth were present is unknown.

In general terms, provided the severity of the wear does not result in short clinical crowns, the prosthesis can be made with conventional techniques. However, if short clinical crowns are present the complexity of care increases significantly.

Speech

Missing maxillary incisor teeth often lead to difficulties with speech, especially those with lip to tooth and tongue to tooth sounds. However, for most this is a temporary condition which slowly improves as adaption occurs. However, for a small minority the difficulties continue and contribute to challenges in their social interaction. Whilst missing anterior teeth have the greatest potential impact on speech, the over-riding reason to replace the teeth will be, for most patients, the impact on appearance. Patients with congenital or acquired dental and skeletal defects (e.g. cleft lip and palate) and oncology patients require special consideration as it is not only the loss of teeth that impacts upon speech.

Psychological/quality of life

It can be seen from the preceding sections that the loss of teeth can have an impact on a patient's quality of life, particularly those teeth within the smile line, leading to impact on their appearance, lack of confidence and social withdrawal, and in some instances loss of function. The loss of a tooth or teeth is an emotional event for many patients, seen as a sign of old age and takes time to adapt to. The decision on tooth replacement should be driven by the patient and not the dentist. Patients should be informed of the disadvantages associated with the tooth's prosthodontic replacement and so are fully able to give informed consent.

HOW TO REPLACE MISSING TEETH?

The question of 'How to replace missing teeth?' can only be answered on an individual patient basis, taking into consideration the patient's expectations and each clinical situation and presentation.

Removable prostheses

Removable prostheses have a number of advantages over fixed prostheses and can be used in situations that are unsuitable for a fixed alternative. In general, the provision of removable prostheses is less invasive and therefore conservative of tooth tissue and is often quicker and cheaper to provide, especially acrylic-based prostheses. Whilst a removable prosthesis potentially might be seen as an opportunity for patients to thoroughly clean their appliance and teeth, studies have shown that removable prostheses lead to deterioration in oral hygiene, not only in the arch in which the appliance is placed but also in the opposing arch. The distribution of plaque accumulation also changes as the potential plaque stagnation areas within the mouth are altered by the appliance (Figure 17.4). As this will predispose the patient to periodontal problems and caries, careful attention therefore needs to be given to prevention of dental disease.



Figure 17.4 The wearing of this upper removable partial cobalt chromium prosthesis may have contributed to this patient's poor oral hygiene and change in the distribution of plaque accumulation on the proximal tooth surfaces.

Multiple edentulous spaces, long spans and free-end saddles

Removable prostheses are more likely to be considered where there are multiple edentulous spaces, for example as seen in Figure 17.5. It could be argued that each individual edentulous span could be restored with a fixed prosthesis (bridge) if care was taken with the occlusion. However, this would require extensive operative intervention and dismantling of existing crown work, with concomitant risks to the abutment teeth. A removable prosthesis allows all edentulous spaces to be restored with a cobalt chromium denture that can be removed for ease of maintenance.



Figure 17.5 Multiple edentulous spaces and long edentulous spans make this patient unsuitable for fixed bridges. Removable cobalt chromium prostheses are a preferred option.

Where long edentulous spans are concerned, conventional fixed bridges may be contraindicated due to the excessive loads applied to the abutment teeth (see Chapter 19) and increased demands on retention for the retainers. The length of the span also increases the difficulty of achieving parallel preparations on the abutment teeth, increasing the risk of creating undercuts. This would pose a problem if fixed–fixed bridge designs were planned. Consider the edentulous spans in the upper right and lower left quadrants for the patient in Figure 17.5. In this situation removable prostheses are preferred where support for the occlusal load and retention can be obtained not only from the adjacent teeth, but also other teeth in the same arch and mucosa.

A unique situation occurs when there is a single or bilateral free-end saddle (Figure 17.6). In these circumstances the only fixed prostheses possible to replace all or the majority of teeth would be implant retained restorations (Figure 17.7). Surgical placement of implant fixtures can be complicated by the presence of anatomical features such as the inferior dento-alveolar neurovascular bundle in the lower arch and the maxillary sinus in the upper arch. Distal cantilever conventional or minimal preparation (resin retained) bridges are limited in most situations to a single premolar unit replacement (Figure 17.8). As such, the restoration of choice is most commonly a removable partial prosthesis.

Ridge resorption

Following an extraction, alveolar ridge resorption takes place. Therefore, if a fixed prosthesis is planned, it is important that the definitive restoration is delayed until most of the resorption has occurred. Whilst ridge resorption following tooth loss continues over a lifetime, most resorption occurs during the first 3 months. The rate of resorption slows but continues over many years; however, the rate and extent of resorption over time will depend on numerous factors, such as:

- The anatomy of the residual ridges, including the quantity and quality of bone (e.g. bone density)



Figure 17.6 Bilateral free end saddles. Treatment options include a removable prosthesis, implant retained restorations or distal cantilever bridges off the premolar teeth. The last option would only allow replacement of a single premolar unit and could be minimum preparation (resin retained) in design on the patient's right and conventional on the left.
(Courtesy of Dr Brendan Scott)



Figure 17.7 Loss of teeth in the upper right quadrant has led to an extensive unilateral free-end saddle. The only fixed prosthesis possible is an implant retained bridge. Care in treatment planning and implant placement is required to avoid the maxillary sinus.

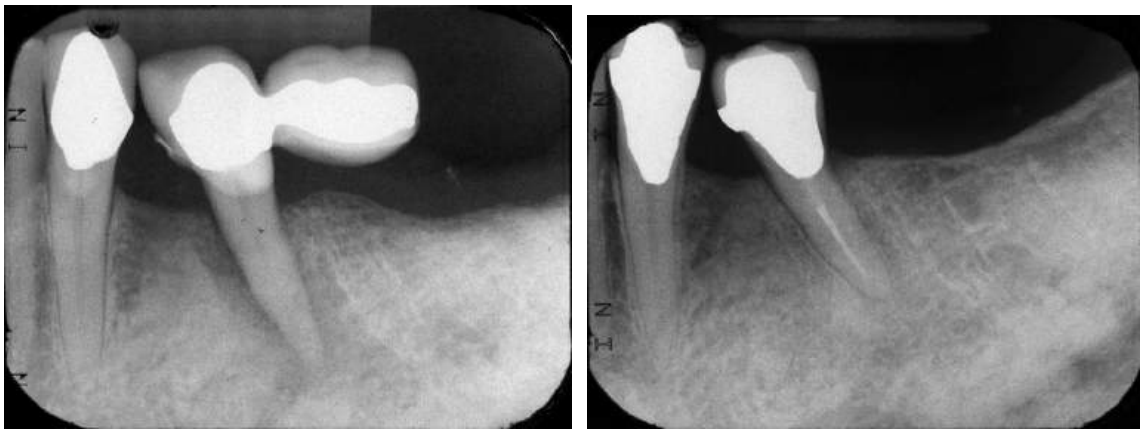


Figure 17.8 A molar pontic has been cantilevered off this premolar tooth. Excessive occlusal forces together with loss of vitality of this tooth have led to increase mobility and bone loss (left). Root canal treatment and removal of the pontic has resolved these problems (right).

- Advanced periodontal disease resulting in severe alveolar bone resorption
- Metabolic factors: for example, the imbalance in sex hormones can result in an imbalance in bone resorption and bone formation, leading to osteoporosis and more rapid alveolar resorption
- Functional forces: it is not uncommon when heavy loads are applied to an edentulous ridge to see a marked increase in resorption leading to atrophy. Such a situation can be seen when a mucosal-borne removable prosthesis is opposed by a natural dentition. The excessive forces generated by the natural teeth to the opposing edentulous ridge via the prosthesis are sufficient to cause marked ridge resorption (Figure 17.9).

Where bridges are planned

It is generally accepted that the rate of ridge resorption following an extraction will have slowed sufficiently after 6 months to allow placement of a fixed prosthesis (bridge). Placement of a fixed prosthesis before this runs the risk of continued ridge resorption beneath the bridge pontic and formation of an unsightly gap.

Where implants are planned

The timing of implant fixture placement is influenced by the health of the bone surrounding the planned site. Sufficient time is needed for resolution of any infection or tissue damage, and soft tissue and bony healing prior to implant placement. Where there is no infection and stability of the implant can be achieved, an immediate implant placement into the extraction socket is possible. There is little evidence on the long-term outcomes of immediately placed implants compared to delayed implant placement following extraction. Careful case selection is required with immediate implant placement and compared to delayed conventional implant placement there would appear to be a higher risk of



Figure 17.9 A natural dentition in the upper arch opposing a complete removable prosthesis can lead to marked alveolar ridge resorption in the lower arch over time.
(Courtesy of Dr Brendan Scott)

failure (Schropp & Isidor, 2008). Immediate loading of implants has also been suggested but this takes a great deal of planning and again the risk of failure is likely to be higher.

Whether implant placement is immediate or delayed a provisional restoration is needed in the majority of cases; if a removable prosthesis is used as a provisional tooth replacement, excessive loading of the implant site immediately after placement should be avoided. This is particularly true if a single-stage surgical procedure is carried out and the transmucosal abutment is visible.

Immediate prostheses

The extraction of anterior teeth and the immediate replacement with a removable prosthesis eliminates some of the aesthetic problems associated with tooth loss. This clinical technique is relatively straightforward for single tooth replacement but becomes more challenging as the number of teeth extracted increases. An impression is taken of both dental arches prior to the tooth extraction, the technician then removes the tooth that is to be extracted and associated undercuts from the resultant working model, and an acrylic prosthesis is processed. Once the tooth has been extracted the denture is fitted immediately into the edentulous space, avoiding an unsightly gap for any time (Figure 17.10).

Severe alveolar bone loss

Occasionally there can be marked bone loss following tooth removal. This may be due to a number of reasons such as the adverse factors previously mentioned, significant preoperative periradicular pathology, following surgical extraction and bone removal (Figure 17.11) or where teeth have been lost through trauma with associated loss of alveolar bone. Other patients may have a deficit of bone if teeth are congenitally missing, if there is a developmental defect such as a cleft palate or if surgery has been required for removal of pathology (Figure 17.12). In these patients not only can the teeth be replaced with a removable prosthesis, but the lost bone and soft tissue can also be replaced with an acrylic flange or obturator.

Single missing teeth with spacing

Some patients have a spaced dentition simply due to a dento-skeletal discrepancy (Figure 17.13). Anterior tooth replacement in such patients poses a particular problem. To place a tooth of equal width to the contralateral counterpart, in the middle of a larger space, can only be achieved with a removable prosthesis (see Figure 17.10 with immediate denture) or an implant (Figure 17.14). Historically, spring cantilever bridges were used but these are now obsolete (Figure 17.15).

Transitional prostheses

Where teeth are lost and there are other teeth of a poor prognosis within the dentition, fixed prostheses are usually contraindicated and a removable acrylic prosthesis can be used as a transitional prosthesis. This can be designed so that it is easy to add any further teeth lost in the future.

Fixed prostheses

For most patients a fixed solution to replace missing teeth is preferred. This might be due to the fact there is no palatal or lingual connector or that they are smaller and do not cover extensive areas of alveolar ridge. Some patients may simply not be able to tolerate a removable prosthesis





Figure 17.10 The upper left central incisor has suffered a root fracture and requires extraction. An immediate removable acrylic prosthesis has been made and fitted immediately after the tooth was extracted, avoiding the patient having an unsightly space for any period of time.



Figure 17.11 This patient has had an upper left canine surgically removed leaving a marked alveolar defect (top). To avoid an unsightly relationship between the bridge pontic planned and the edentulous ridge, an alveolar bone graft has been performed (bottom).

due to a retch reflex for example. A fixed restoration gives patients added security and psychologically becomes an integral part of the dentition but generally increases cost. There are two types of fixed restoration that can be considered: bridges and implants.

Bridges

There are two main types of bridges available to the dentist: conventional bridges and minimal preparation (resin retained) bridges. Irrespective of the type of bridge, both of which will be described in detail in the subsequent two chapters, the terminology used to describe the components is common to both.

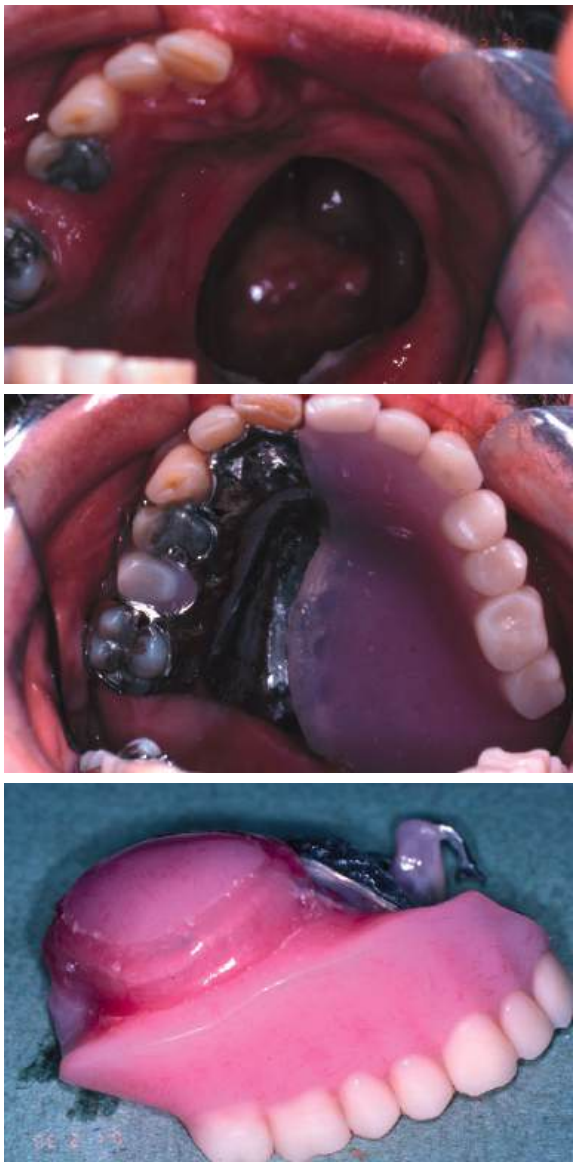


Figure 17.12 This patient has had surgery to remove a tumour leaving a large defect in the maxilla. The removable prosthesis not only replaces the missing teeth, but it also replaces the lost tissues and seals the oro-nasal communication.

Conventionally prepared bridges

Conventional bridges are usually indicated when the abutment teeth are extensively restored, requiring preparation of the teeth for conventional retainers, normally full coverage crowns. Full coverage retainers protect the remaining coronal tooth tissue of the abutment teeth and the core material from occlusal loading, and provide optimum retention for the bridge.



Figure 17.13 Patient with dento-skeletal discrepancy leading to spacing. Loss of an anterior tooth will require a removable prosthesis or implant retained restoration to produce the same appearance.

Minimal preparation (resin retained) bridges

Minimally prepared (resin retained) bridges are indicated when the abutment teeth are unrestored or minimally restored. Rather than preparing virgin teeth for full coverage retainers, the abutment teeth are minimally prepared and the primary retention for the bridge is achieved from adhesive cements (see Chapter 18). It could be argued that with improved dental health and fewer heavily restored teeth, minimally prepared (resin retained) bridges are, or may become, the more 'conventional' of the bridge designs. However, with the rapidly growing market in implants, these too may become the restoration of choice for the minimally restored dentition in the future.

Implant retained restorations

Both conventional and minimally prepared (resin retained) bridges use adjacent teeth to gain support and retention for the prosthetic tooth or pontic. As such, caries



Figure 17.14 A missing central incisor tooth initially replaced by a cobalt chromium removable prosthesis and latterly by a single tooth implant. Both treatment options preserve the midline diastema.



Figure 17.15 This patient had a midline diastema which has been preserved by placement of a spring cantilever bridge. A single tooth implant would be the preferred treatment option today.

can occur at the margins of the restoration and periodontal problems can arise in susceptible patients (Figure 17.16). In most situations this risk can be reduced by effective preventive management given preoperatively and postoperatively and throughout any reviews and maintenance. In some situations the adjacent teeth are unsuitable for bridge abutments by virtue of the tooth size (crown and/or root) and periodontal support compared to the length of the edentulous span. Such situations can be treated with implant retained crowns, bridges or dentures (Figure 17.17).



Figure 17.16 The resin retained bridge on the upper left premolar tooth has failed due to secondary caries.

Orthodontic treatment and replacing teeth

Whenever a patient is provided with a restoration of any kind, and in particular costly indirect restorations, it is important to emphasize that they will require maintenance and possible repair or replacement throughout life. For some this prospect is not acceptable and the spaces can be closed orthodontically. A typical example is the management of congenitally missing upper lateral incisor teeth. A decision has to be made as to whether to close the space orthodontically or to open up sufficient space to place prosthetic lateral incisor teeth. The former avoids prosthetic replacement but the appearance may be compromised by darker coloured, pointed and bulbous canine teeth adjacent to the central incisor teeth. Whilst the shape of the canine teeth can be altered to some degree by addition of composite resin and judicious tooth adjustment, gingival contour, canine eminence and lack of canine guidance cannot be overcome (Figure 17.18).

A superior appearance can be obtained by creating space for a prosthetic tooth (Figure 17.19); however, this may involve extraction of a sound tooth more posterior to the canine tooth, providing space for distal tooth movement. It also relies on a co-operative patient who is prepared to undergo a more protracted course of orthodontic treatment and accept a prosthetic replacement and its maintenance throughout life.



Figure 17.17 Replacement of upper right central incisor with an implant retained crown in a patient with an unrestored dentition and no periodontal disease.



Figure 17.18 The upper right lateral incisor is congenitally missing, the upper left is diminutive in size and there is a midline diastema. As the patient did not want orthodontic treatment the upper anterior teeth have been restored with composite. The result is a reasonable compromise.



Figure 17.19 Patient with congenitally missing upper left lateral incisor and diminutive right lateral incisor tooth. Orthodontic treatment has produced sufficient space for a resin retained bridge on the left and a composite build-up on the right.

(Continued)



Figure 17.19—cont'd

SUMMARY

When a patient loses or has a missing tooth or teeth, their replacement needs to be considered carefully in light of the advantages and disadvantages of doing so. Once a decision to replace missing teeth has been made, each clinical situation needs to be evaluated carefully with the aid of appropriate special investigations such as radiographs and articulated study models. The ways of how to replace the teeth should then be discussed, paying particular attention to the prognosis of each.

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